

Does the Design Meet California's Bay Discharge Rule? Discharge Limits, Fan Heat, Fouling, Condensation and Recirculation

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11.3 K

discharge rise with the old fixed pump flow (rule: 2.2 K)

41×

less Bay water than a rule-compliant seawater-first design

187 vs 201

MWh/yr: hybrid (sized + bypass) vs seawater first

0.82 MGD

design intake, under the 2 MGD federal threshold

1. Summary

California's Thermal Plan names San Francisco Bay as an enclosed bay and prohibits new cooling-water discharges more than 4 °F (2.2 K) above the natural water temperature. The earlier models pumped a fixed 3.25 kg/s of seawater per path. That flow returns water up to 11.3 K warmer than the Bay, above the limit in 26% of pumping hours. The seawater-first baseline breaks it in every hour.

Result: a variable-speed pump that holds the rise at 2.0 K fixes this. The hybrid's yearly seawater use barely changes (27,400 m³ instead of 30,500 m³), but the pump, intake and pipes must be sized for about 18 kg/s per path, roughly 5.6 times the old design flow. The seawater-first baseline is hit much harder: to stay within the limit it must pump about 1.13 million m³ a year, not 200,000 m³. That costs it about 45 MWh of extra pump energy, so it uses 201 MWh a year against 187 MWh for the hybrid with the sized exchanger and coil bypass. Once both designs follow the discharge rule, the energy premium reported in paper v5 disappears. The two designs use about the same energy, and the hybrid pumps about 41 times less Bay water.

Five further checks need no new CAD or CFD: fan heat, exchanger fouling, condensation, exhaust recirculation between modules, and the effectiveness of commercial exchanger cores. None of them changes the conclusion. Two set firm design requirements: the Stage 1 exchanger must reach about 0.75 effectiveness, and intake and exhaust must be kept apart.

2. The Discharge Rule

The State Water Resources Control Board's Water Quality Control Plan for Control of Temperature in the Coastal and Interstate Waters and Enclosed Bays and Estuaries of California (the Thermal Plan) lists San Francisco Bay among enclosed bays. For new discharges to enclosed bays it states: "Thermal waste discharges having a maximum temperature greater than 4 °F above the natural temperature of the receiving water are prohibited." Cooling water used to carry away waste heat is thermal waste under the Plan's definitions.

4 °F is 2.22 K. The models below use a 2.0 K design cap to leave a margin for sensor error. The rule applies to the maximum discharge temperature, not to an annual average, so the design must meet it in every hour.

3. What the Fixed-Flow Design Did

Design	Max discharge rise	Mean rise when pumping	Pumping hours over 2.22 K
Hybrid, fixed 3.25 kg/s per path	11.3 K	1.8 K	26%
Seawater first, fixed 3.25 kg/s per path	11.3 K	11.3 K	100%

The hybrid's average rise was already low, because most pumping hours only trim a little heat. On hot afternoons, though, the coil takes up to about 146 kW per path. At 3.25 kg/s that heats the water by 11 K. Neither design as modelled in paper v5 could be permitted.

4. A Rule-Compliant Seawater Loop

The seawater coil is now modelled with a variable-speed pump. In each hour the pump runs at the lowest flow that both brings the air to the setpoint and keeps the water rise at or below 2.0 K. The coil's conductance is fixed at its original design value, so more water flow also raises its effectiveness. Pipes are sized for the peak flow: 50 kPa of static lift plus friction that scales with flow squared and reaches 160 kPa at peak.

Case	Pump hours	Peak flow / path	Bay water / yr	Heat to Bay	Pump energy
Hybrid, 21 °C supply	15.2%	18.3 kg/s	27,400 m ³	62 MWh	0.9 MWh
Hybrid, 24 °C supply*	4.4%	18.3 kg/s	8,900 m ³	20 MWh	0.3 MWh
Seawater first, 21 °C supply	100%	23.9 kg/s	1,128,000 m ³	2,563 MWh	59.8 MWh

* 24 °C assumes return air at 36.25 °C, pending CFD Test 3.

Capping the temperature rise does not change the hybrid's yearly water use much, because water use follows heat moved divided by the rise. It does set the size of the hardware: about 18 kg/s per path (about 280 GPM), against the 3.25 kg/s used before. The intake, strainer, pump and pipes in the CAD must be sized for that flow.

5. Energy With the Rule Applied

Design (all rule-compliant)	Cooling electricity	Mech. PUE	Bay water / yr
Seawater first	201 MWh/yr	1.08	1,128,000 m ³
Hybrid, assumed 200 Pa exchanger	273 MWh/yr	1.12	27,400 m ³
Hybrid, 200 Pa + coil bypass	243 MWh/yr	1.10	27,400 m ³
Hybrid, sized 67 Pa exchanger	217 MWh/yr	1.09	27,400 m ³
Hybrid, sized 67 Pa + coil bypass	187 MWh/yr	1.08	27,400 m ³

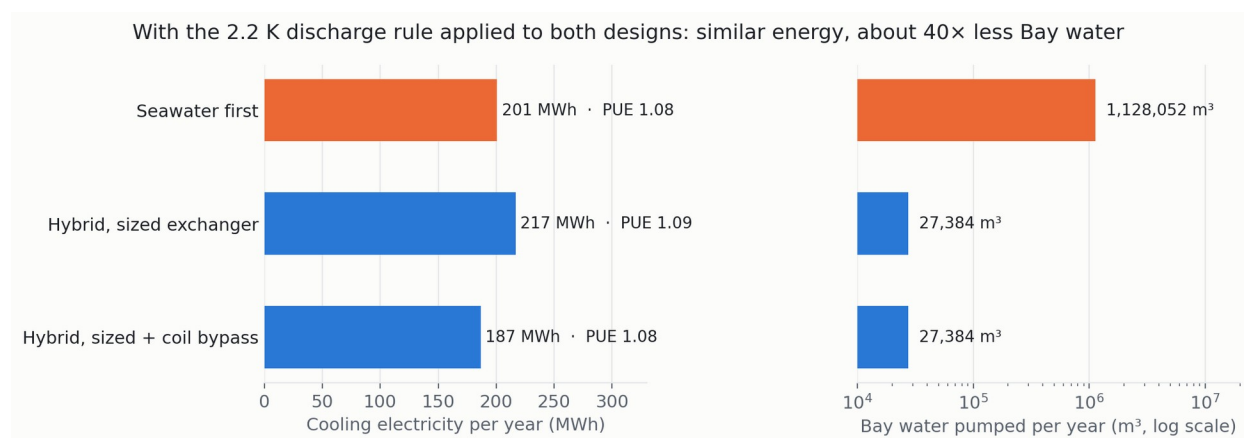


Figure 1. Annual cooling electricity and Bay water pumped when both designs keep the discharge within 2 K of Bay temperature.

Paper v5 found that pumping Bay water every hour used less energy than the hybrid (157 MWh against 188–218 MWh). That comparison let the seawater-first design break the discharge rule. Once both designs comply, seawater first needs about 5.6 times more water flow, and its pump energy rises from about 15 to 60 MWh a year. The hybrid with the sized exchanger and coil bypass then uses slightly less energy (187 against 201 MWh). Without the bypass it uses slightly more (217 MWh).

This comparison depends on the seawater pipe friction, which is an assumption. With a 100 kPa design head (larger pipes), seawater first falls to 182 MWh; with 220 kPa it rises to 219 MWh. The hybrid barely moves, because its pump runs only 15% of the time. The fair conclusion is energy parity within about $\pm 10\%$, with the hybrid pumping about 41 times less Bay water and returning about 41 times less heat.

6. Intake Regulations

Federal Clean Water Act Section 316(b) rules for new facilities (40 CFR 125.81) apply when the design intake flow exceeds 2 million gallons per day (MGD) and at least 25% of the water is used for cooling. The threshold is set by design flow, not by how often the pump runs.

- 270 kW module: peak intake $0.036 \text{ m}^3/\text{s} = 0.82 \text{ MGD}$, below the threshold.
- 1 MW (3.7 modules): about **3.0 MGD**, above it. Seawater first is above it at a similar size, because both designs need a similar peak flow.
- Screen area for the 0.5 ft/s (0.15 m/s) through-screen velocity limit: about 0.24 m^2 of open area at 270 kW, which is small and easy to fit.

So the hybrid does not avoid intake regulation at facility scale through its design flow. Its regulatory advantage is the yearly volume withdrawn, about 41 times less, which sets entrainment of larvae and plankton, and the heat returned to the Bay.

7. Robustness Checks

7.1 Fan heat

All of the room fans' electrical power ends up as heat in the room-air loop: 18.4 kW with the sized exchanger, or 22.8 kW with the assumed 200 Pa unit. With 270 kW of IT, the loop must remove 288–293 kW. The model already removes 292.6 kW, because it uses the CFD return temperature (12.25 K rise at 10.06 m³/s per path). Fan heat is therefore already covered, and the 21 °C results do not change.

7.2 Fouling of the outdoor side

Salt and dust collect on the outdoor-air side of the plates. In these laminar channels the thermal penalty is tiny: the deposit narrows the channel, which raises the heat-transfer coefficient about as much as the deposit's own resistance lowers it. The real penalty is pressure drop, which scales with one over the gap cubed.

Design	Deposit per wall	Outdoor-side drop	Effectiveness	Cooling energy (bypass)
3 mm gap, 1.2 m	clean	67 Pa	0.75	188 MWh
3 mm gap, 1.2 m	0.25 mm	115 Pa	0.76	193 MWh
3 mm gap, 1.2 m	0.5 mm	224 Pa	0.77	204 MWh
4 mm gap, 1.5 m	clean	33 Pa	0.75	174 MWh
4 mm gap, 1.5 m	0.5 mm	78 Pa	0.76	177 MWh

A 3 mm gap with 0.5 mm of deposit per wall triples the outdoor-side pressure drop but adds only about 16 MWh a year. A 4 mm gap is far more tolerant and gives the lowest energy, at the cost of a 3 m tall stack. The 4 mm gap is the better choice for salt air if deck height allows, and either way the outdoor side needs a washable design and a cleaning interval. Deposit conductivity (0.5 W/m·K) and thickness are assumed values.

7.3 Condensation

The room-air loop is closed, so its dew point is whatever the humidity control holds. Water condenses where a surface is colder than that dew point. The coldest Stage 1 plate corner is taken as halfway between outside air and the 21 °C supply; the seawater coil surface is taken as the Bay water temperature.

Room dew point	Stage 1 plate hours / yr	Seawater coil hours / yr
9 °C	0	0
12 °C	0	2
13 °C	22	36
15 °C (ASHRAE upper recommended)	918	196

If the room dew point is held at about 12 °C or below (about 56% relative humidity at 21 °C supply), condensation is essentially absent. That is inside the ASHRAE recommended range. A drain pan on the seawater coil is still prudent.

7.4 Exhaust recirculation between modules

When modules sit side by side, warm exhaust from one can reach a neighbour’s intake. Finding the actual temperature rise needs a site-scale CFD model; here the rise is applied to every hour to show how much it matters.

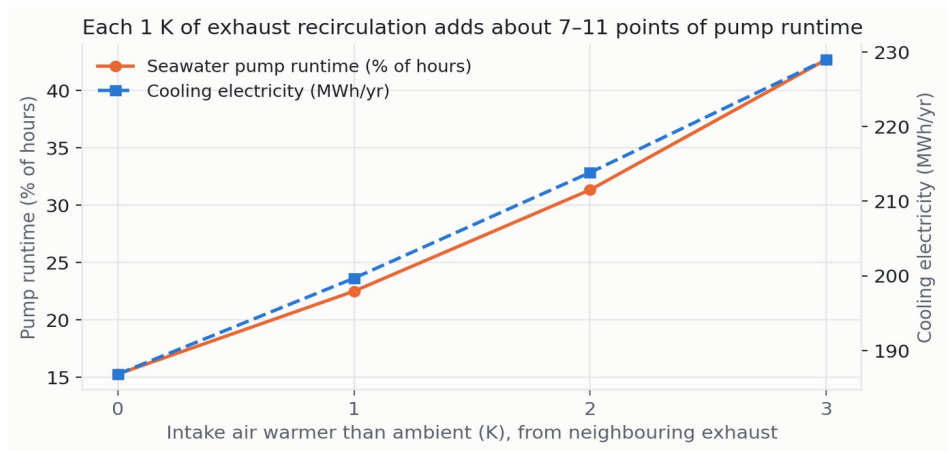


Figure 2. Effect of a uniform intake temperature rise (sized exchanger, coil bypass, rule-compliant pump).

Intake rise	Breeze-only hours	Pump runtime	Cooling energy	Bay water / yr
0 K	84.8%	15.2%	187 MWh	27,400 m³
1 K	77.5%	22.5%	200 MWh	40,400 m³
2 K	68.7%	31.3%	214 MWh	59,000 m³
3 K	57.3%	42.7%	229 MWh	84,600 m³

Recirculation is the most sensitive layout effect found so far. Each kelvin of intake warming adds 7–11 points of pump runtime. Even at 3 K the hybrid still pumps about 13 times less water than seawater first, but the multi-module layout should keep intakes upwind and below exhausts, with exhaust discharged vertically. A 1 MW site-layout CFD study is the right next model.

7.5 Commercial exchanger cores

Commercial indirect cores report dry air-to-air efficiencies from about 60% (a commercial data center unit studied by Wei et al., 2024) up to about 80% (Heatex product literature). Vendors publish no pressure-drop data at operating flow; selection is done in their software, so the 200 Pa assumption still cannot be checked against a datasheet.

Stage 1 effectiveness	Breeze-only hours	Pump runtime	Energy, 200 Pa	Energy, 67 Pa
0.60	44.7%	55.3%	311 MWh	242 MWh
0.75 (design)	84.8%	15.2%	243 MWh	187 MWh
0.80	89.7%	10.3%	231 MWh	177 MWh

Effectiveness is the single most important equipment specification. At 0.60 the breeze covers fewer than half the hours and pumping triples. The design must specify a core rated at 0.75 or better at equal airflows, and the procurement step should get a vendor selection printout that confirms both effectiveness and pressure drop.

8. What Changes in the Paper

- **The headline trade-off changes.** Under the discharge rule the hybrid and seawater first use about the same energy, and the hybrid pumps about 41 times less Bay water. Before, it was 85% less water at an energy premium.
- The seawater loop must be variable-speed and sized for about 18 kg/s per path, not 3.25 kg/s. The CAD intake, pipes and pump need resizing.
- The intake is below the federal 316(b) threshold for one 270 kW module but above it at 1 MW, for both designs.
- Design requirements: Stage 1 effectiveness of at least 0.75, outdoor gap of 4 mm preferred for fouling, room dew point at or below about 12 °C, and intake and exhaust kept apart.

9. Limitations

- The 2.0 K cap is applied to the outlet of the coil. The regional board may also set mixing-zone, receiving-water and biological conditions in a permit; these are not modelled.
- Pump head, pipe friction and deposit properties are assumed values. Seawater-first energy is sensitive to pipe friction (182–219 MWh).
- Recirculation is applied as a uniform rise in every hour; real recirculation depends on wind direction and layout.
- Commercial effectiveness figures come from one published study and manufacturer literature, not from a selection for this duty.

Sources

[1] State Water Resources Control Board, Water Quality Control Plan for Control of Temperature in the Coastal and Interstate Waters and Enclosed Bays and Estuaries of California (Thermal Plan). Definitions and Specific Water Quality Objectives for Enclosed Bays, new discharges.

[2] 40 CFR 125.81, Who is subject to this subpart (new facilities, design intake flow greater than 2 MGD), and 40 CFR 125.84 (through-screen velocity 0.5 ft/s).

[3] Q. Wei, J. Lu, X. Xia, B. Zhang, X. Ying and L. Li, “Performance and applicability analysis of indirect evaporative cooling units in data centers across various humidity regions,” *Buildings*, vol. 14, no. 11, art. 3623, 2024.

[4] Heatex, Model H2 cross-flow plate heat exchanger product data sheet and Indirect Evaporative Cooling brochure (efficiency up to 80%; selection via Heatex Select software).

[5] ASHRAE Technical Committee 9.9, Thermal Guidelines for Data Processing Environments, 5th ed., 2021.

[6] R. K. Shah and A. L. London, *Laminar Flow Forced Convection in Ducts*, Academic Press, 1978.

Code and data: github.com/sharanankathi/bay-breeze-cooling (bay_discharge.py, robustness.py, plots_discharge.py).